

SMART RINGS

What are Smart Rings?

A smart ring is a compact wearable device that typically fits on a finger and functions similarly to a smartwatch. It works by integrating small electronic components such as sensors, microcontrollers, and wireless modules into a ring-shaped form factor.

Components in a Smart Ring

Sensors
Power Components
Communication Module
Microcontroller/Microprocessor
Enclosure (could be plastic, titanium etc.)
Storage (optional, a form of flash memory)

Types of Sensors

PPG Sensor (Photoplethysmography)

Skin Temperature Sensor

EDA Sensor (Electrodermal Activity)

Accelerometer

SpO2 Sensor

Proximity or Capacitive Sensor

Communication Module

Bluetooth Low Energy

NFC chip (some rings)

SENSORS

PPG Sensor

What is a PPG Sensor?

A PPG (Photoplethysmography) sensor works by using light to measure changes in blood volume in the microvascular tissue beneath the skin. It's the core sensor in most smart rings, watches, and fitness trackers for measuring heart rate, heart rate variability (HRV), blood oxygen levels (SpO₂), and sometimes respiratory rate.

How does a PPG Sensor work?

Light Emission

- The sensor emits light (usually green, red, or infrared) into your skin.
- Green light is absorbed by blood more strongly than surrounding tissue, especially because hemoglobin absorbs green light well.

Light Absorption and Reflection

- As the heart pumps blood, the volume of blood in the skin rises and falls with each heartbeat.
- This causes changes in how much light is absorbed and reflected.

Photodetector Receives Reflected Light

- A photodiode (light detector) next to the LED captures the amount of reflected or transmitted light.
- More blood = more absorption = less light reflected back.
- Less blood = less absorption = more light reflected back.

Signal Processing

- The sensor system converts these light fluctuations into an electrical signal.
- This signal is a pulse waveform—each peak corresponds to a heartbeat.

Analysis by Microcontroller

The device's processor interprets the waveform to calculate:

- Heart Rate (beats per minute)
- HRV (variation in time between heartbeats)
- SpO₂ (if red and IR light are used)
- Respiratory Rate (from HR and waveform variability)
- Sleep Stages (inferred from HR and movement)

Note:

Reflective PPG: Used in smart rings and watches. Light is reflected back from skin and measured.

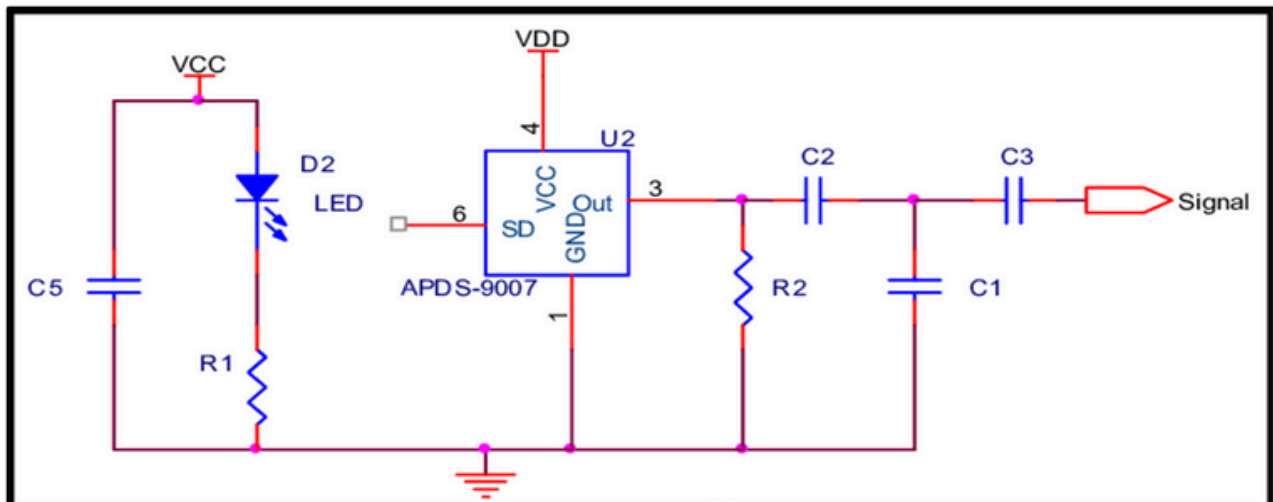
Transmissive PPG: Light passes through tissue (e.g., fingertip or earlobe) and is detected on the other side (used in clinical pulse oximeters).

Why PPG in Smart Rings?

Smart rings use infrared PPG instead of green light because:

- IR penetrates deeper and works better at night (during sleep).
- It's less affected by skin tone and ambient light.
- It's more energy-efficient in ring-sized devices.

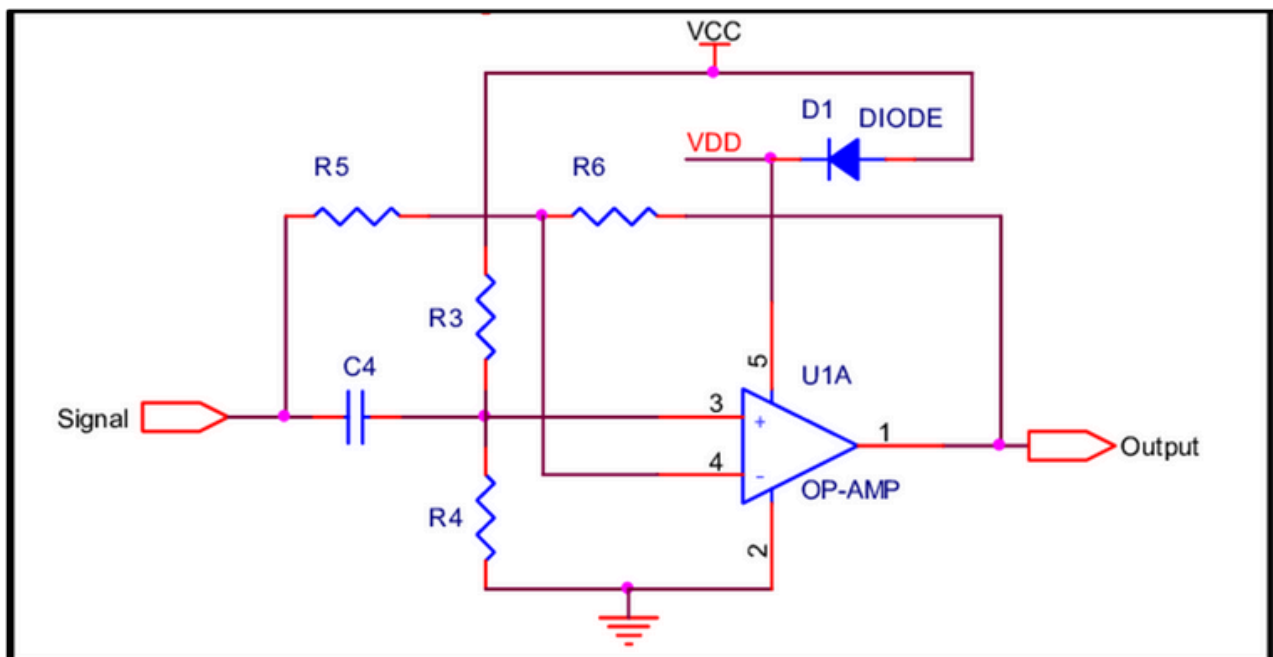
Electronics behind PPG



Light Emission and Detection Circuit

LED & Photodetector (APDS-9007)

- **D2:** Infrared LED - shines light into the skin.
- **R1:** Current-limiting resistor for the LED.
- **U2 (APDS-9007):** Ambient light photodetector (acts like a light-dependent resistor or phototransistor).
- It converts reflected IR light from the skin into a voltage.
- The amount of reflected light changes with blood volume, giving a pulsatile signal (i.e., the PPG waveform).
- **C2, C3, C1, R2:** These act as a high-pass filter and DC blocking network to remove the baseline and pass only the AC component (heartbeats).
- The resulting signal is sent to the Signal output pin for amplification.



Signal Conditioning / Amplification Circuit

Op-Amp Based Amplifier (U1A)

- **U1A:** Operational amplifier (configured as a non-inverting amplifier or active filter).
- **C4:** AC coupling capacitor – blocks DC offset from the previous stage.
- **R3, R4:** Set the gain of the op-amp.
 - $\text{Gain} = 1 + (R3/R4)$
- **R5:** Pull-down or input protection resistor.
- **R6 & D1 (Diode):** Used for clipping or protection, preventing the output from exceeding a certain voltage (e.g., diode clamps at $\sim V_{dd}$).
- **Output:** Conditioned PPG waveform suitable for microcontroller ADC or further processing.

Skin Temperature Sensor

What is a Skin Temperature Sensor?

A skin temperature sensor in a smart ring works by measuring the surface temperature of your skin, usually on your finger, to provide insights into your metabolism, stress, sleep, and overall health.

Most commonly used temperature sensors in smart rings are:

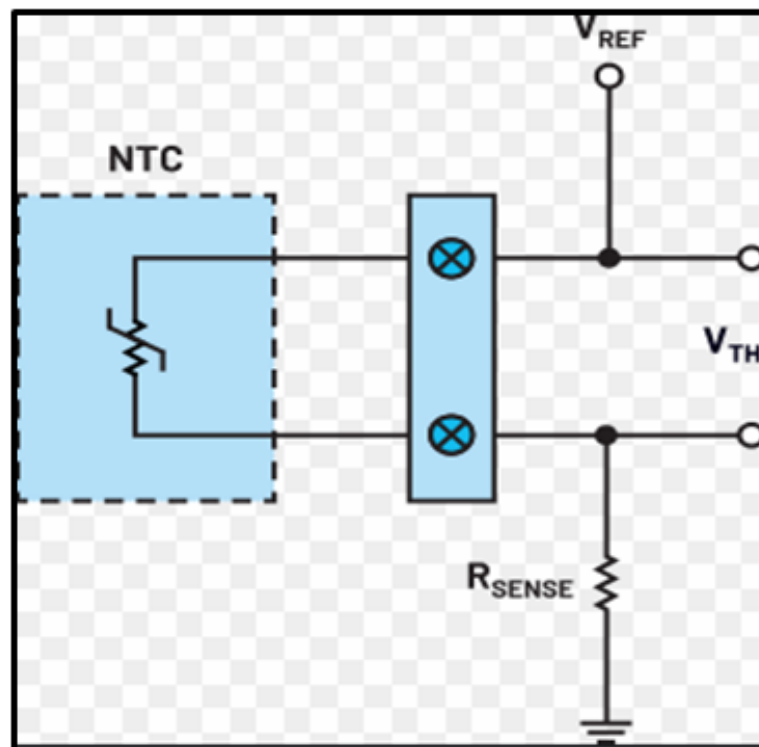
- **Thermistors**
- **Digital Temperature Sensors**
- **Infrared Thermopiles (less commonly used)**

What is a Thermistor?

A thermistor is a type of resistor whose resistance changes with temperature:

- **NTC (Negative Temperature Coefficient):** Resistance decreases as temperature increases.
- **PTC (Positive Temperature Coefficient):** Resistance increases as temperature increases.
- A microcontroller measures the voltage across the thermistor (typically in a voltage divider setup).
- Using calibration, it converts **voltage** → **resistance** → **temperature**.

Electronics behind a Thermistor



Thermistor

NTC Thermistor (Negative Temperature Coefficient):

- Resistance decreases as temperature increases.
- It's the temperature-sensing element.

RSENSE (Fixed Resistor):

- Acts as part of a voltage divider with the thermistor.
- Converts the resistance change of the NTC into a voltage change.

VREF (Reference Voltage):

- A stable DC voltage source applied across the divider circuit.
- Typical values: 3.3V, 5V, etc.

VTH (Thermistor Output Voltage):

- The voltage measured at the junction between the NTC and R_{SENSE} .
- This is the output voltage you feed to an ADC (Analog-to-Digital Converter) to read temperature.

This is a voltage divider configuration:

$$V_{TH} = V_{REF} \times \frac{R_{SENSE}}{R_{SENSE} + R_{NTC}}$$

When temperature increases:

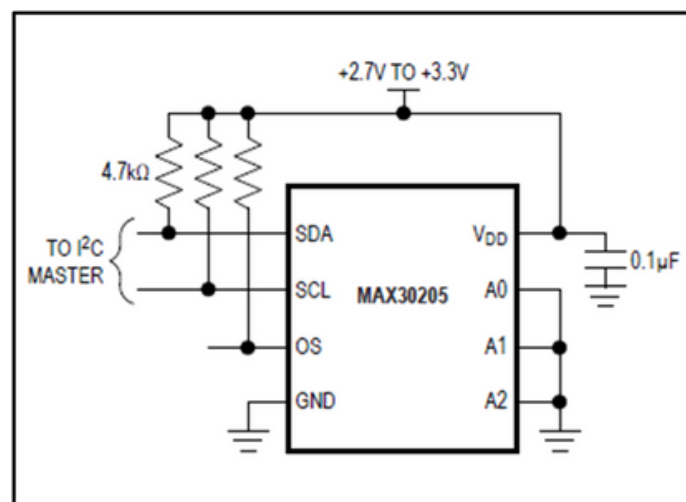
- NTC resistance decreases.
- So RSENSE becomes a larger part of the total resistance.
- Thus, VTH increases.

When temperature decreases:

- NTC resistance increases.
- VTH decreases.

What are Digital Temperature Sensors?

- These are integrated circuits that use precision sensing elements (like bandgap references or silicon diodes).
- They output temperature readings directly in digital format (via I2C or SPI).
- High accuracy (± 0.1 – 0.3°C) and tiny size make them ideal for smart rings.
- E.g., Maxim Integrated MAX30205, Texas Instruments TMP117



Digital Temperature Sensor

Power Supply

- VDD is connected to a 2.7–3.3V power source.
- A 0.1 μF decoupling capacitor is placed close to the power pins to filter noise and stabilize the supply voltage.

I²C Communication

- SDA and SCL lines are connected to the I²C master (typically a microcontroller like an Arduino, STM32, or ESP32).
- 4.7k Ω pull-up resistors ensure the lines are pulled HIGH when idle, as required by the I²C protocol.

Address Selection

- A0, A1, A2 are tied to GND, which sets the device I²C address (you can configure up to 8 unique addresses by varying these).

Temperature Measurement

- The sensor measures temperature internally using a precision analog temperature-sensing element and outputs the value digitally.
- The temperature reading can be retrieved over I²C in 16-bit digital format, accurate to $\pm 0.1^\circ\text{C}$.

EDA Sensor

What is an EDA Sensor?

- An EDA sensor (Electrodermal Activity sensor), also known as a GSR sensor (Galvanic Skin Response), measures the electrical conductance of the skin, which varies with sweat gland activity. It's a widely used biosensor in wearables, including smart rings, for detecting stress, arousal, and emotional states.
- Your skin contains sweat glands that are controlled by the sympathetic nervous system (the “fight or flight” system). When you're stressed, anxious, or excited, even before you start to visibly sweat, the sweat glands activate slightly, increasing the skin's conductance.
 - Dry skin = low conductance (high resistance)
 - Sweaty skin = high conductance (low resistance)
- These changes happen in real-time, making EDA a powerful indicator of psychological or physiological arousal.
- EDA sensors measure the skin conductance between two electrodes placed on the skin, typically the fingers or palm, where sweat glands are dense.

How does an EDA Sensor work?

Contact with Skin

- The smart ring has two electrodes (usually metal pads) on the inner side.

- When the ring is worn, these electrodes maintain contact with the skin, typically on the finger.

Low-Voltage Signal Application

- A very small voltage (in the order of 0.5V or less) is applied across the two electrodes.
- This is done using an analog front-end (AFE) or microcontroller inside the ring.

Measurement of Skin Conductance

- The current that flows through the skin (which acts as a variable resistor) is measured.
- As sweat gland activity increases (due to stress or excitement), skin conductance increases (i.e., resistance decreases).

Signal Conditioning

- The analog signal (voltage drop/current) is often weak and noisy.
- It is amplified, filtered, and passed through analog-to-digital conversion (ADC).

Digital Signal Processing (DSP)

- The digitized signal is processed to remove artifacts and extract meaningful features:
 - Tonic level: Baseline conductance.
 - Phasic response: Rapid changes due to stimuli.
 - EDA peaks: Number and amplitude of responses per minute.

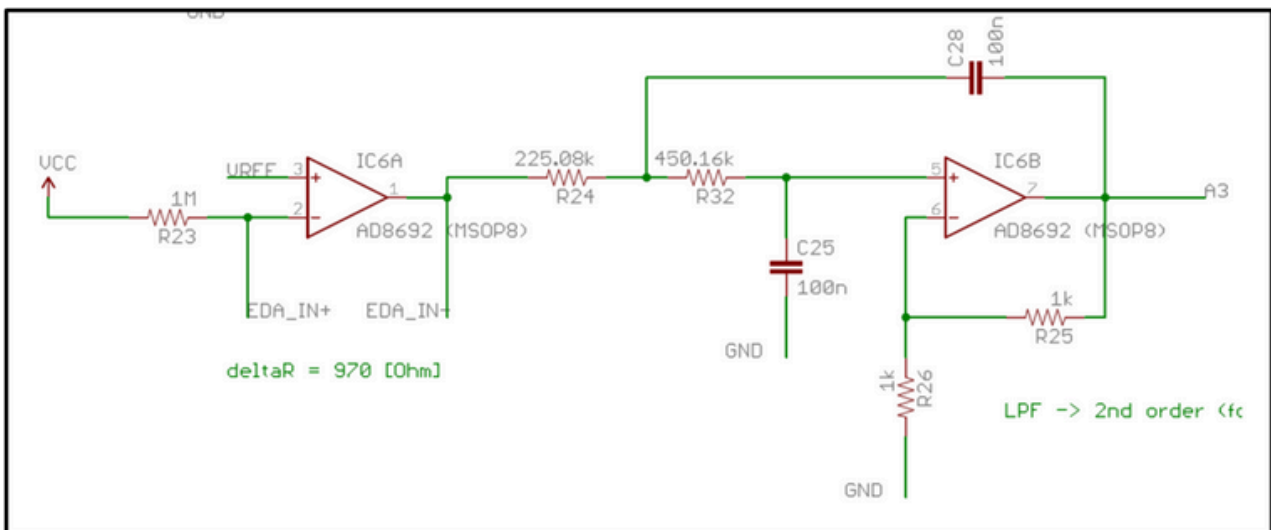
Integration with Other Sensors

- Often combined with:
 - PPG sensors (for heart rate)
 - Temperature sensors
 - Motion sensors (IMU)
- This helps in correlating emotional responses with heart rate or activity level.

Data Transmission

- Processed EDA data is stored locally or sent via Bluetooth to a companion mobile app.
- The app might display:
 - Stress levels
 - Relaxation periods
 - Biofeedback for meditation, etc.

Electronics behind EDA Sensor



Electrodermal Activity Sensor with Adaptive Gain Control

Instrumentation Amplifier (IC6A - MCP6022)

- **Purpose:** Amplifies the small voltage difference between the two electrodes placed on the skin.
- **Why?** The signal from skin is very weak (in microvolts or millivolts), and this stage boosts it.
- **EDA_IN+ and EDA_IN-:** Connected to electrodes touching the skin.
- **R23 (470 kΩ):** Sets the gain of the amplifier (common in differential amplifiers).

Low-Pass Filter (LPF - 2nd Order)

- **IC6B:** Another op-amp stage used as an active low-pass filter.
- **C25 (100 nF), R26 (220 k Ω), R27 (680 k Ω):** Components that define the cutoff frequency of the filter.
- This removes high-frequency noise, keeping only the slow-varying EDA signal (typically under 10 Hz).
- Why LPF? EDA changes slowly with emotional state, so higher-frequency signals are mostly noise.

Voltage Divider and Biasing Network

- **R22, R25, C24:** Provide a reference voltage and decoupling to keep the signal centered within ADC range.
- Also helps in biasing the op-amps properly when working with single supply (like battery).

Accelerometer

What is an Accelerometer?

An accelerometer is a sensor that measures acceleration forces, which can be due to:

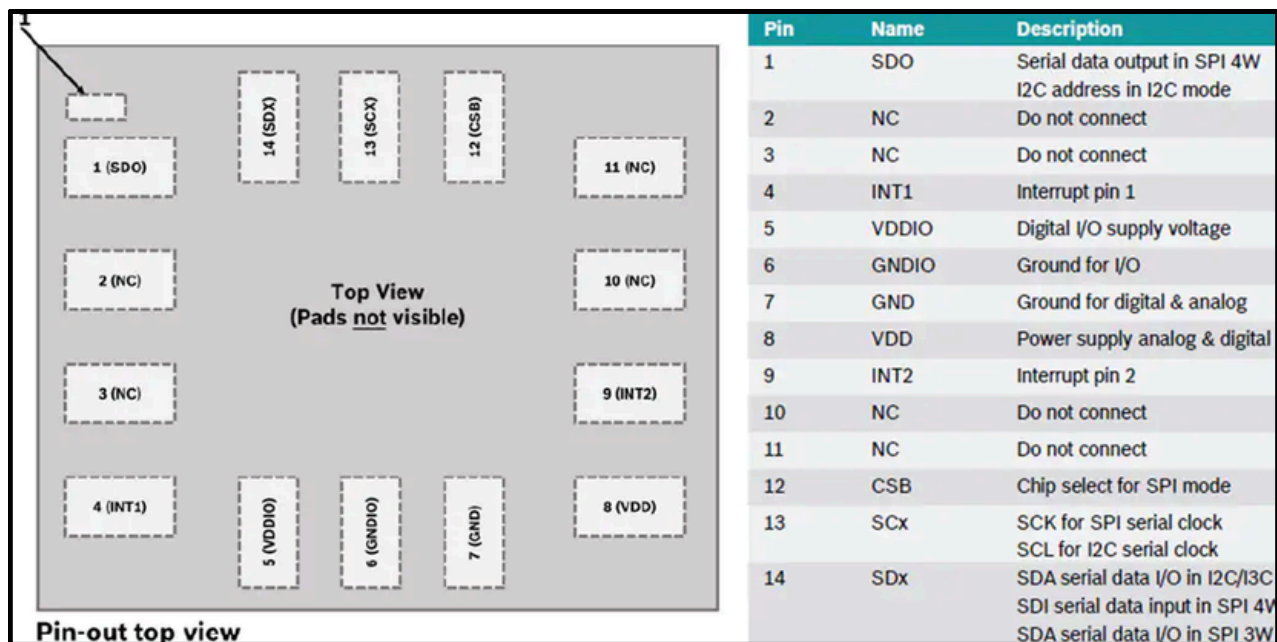
- Movement (dynamic acceleration) – like shaking or tapping
- Gravity (static acceleration) – used to detect tilt and orientation

In a smart ring, the accelerometer is a tiny MEMS (Micro-Electro-Mechanical Systems) device that tracks your motion in 3D (X, Y, Z axes).

How does an Accelerometer work?

- Tiny mechanical structures are suspended in silicon.
- When the ring moves or tilts, these structures shift slightly.
- This displacement changes the capacitance or resistance in the sensor.
- The change is converted into an electrical signal proportional to acceleration.
- A built-in ADC (Analog-to-Digital Converter) outputs digital values for the X, Y, and Z directions.

Electronics behind Accerlerometer



Accelerometer BMI323

An accelerometer/IMU measures motion and orientation:

Accelerometer:

- Detects linear acceleration (X, Y, Z axes).
- Inside, tiny MEMS structures deflect under acceleration, changing capacitance → converted into a digital value.

Gyroscope (if included in IMU):

- Measures angular velocity (rotation).
- Uses vibrating MEMS masses and the Coriolis effect.

Digital Output:

- The chip outputs digital signals (via I²C or SPI) to the microcontroller (MCU) inside the smart ring.
- Interrupts (INT1, INT2) can signal events like a "step detected" or "motion detected."

SpO₂ Sensor

What is an SpO₂ sensor?

- SpO₂ stands for Peripheral Capillary Oxygen Saturation.
- It's a non-invasive optical sensor that measures the percentage of oxygenated hemoglobin (HbO₂) compared to total hemoglobin (Hb) in the blood.
- In simple words → it tells how well oxygen is being carried through your bloodstream.

How does an SpO₂ sensor work?

An SpO₂ sensor uses a method called Pulse Oximetry, which is based on light absorption of blood.

Light Emitters (LEDs):

- Typically two LEDs are used:
- Red Light (~660 nm) → absorbed differently by oxygenated vs. deoxygenated blood.
- Infrared Light (~940 nm) → absorbed mainly by oxygenated blood.

Photodetector:

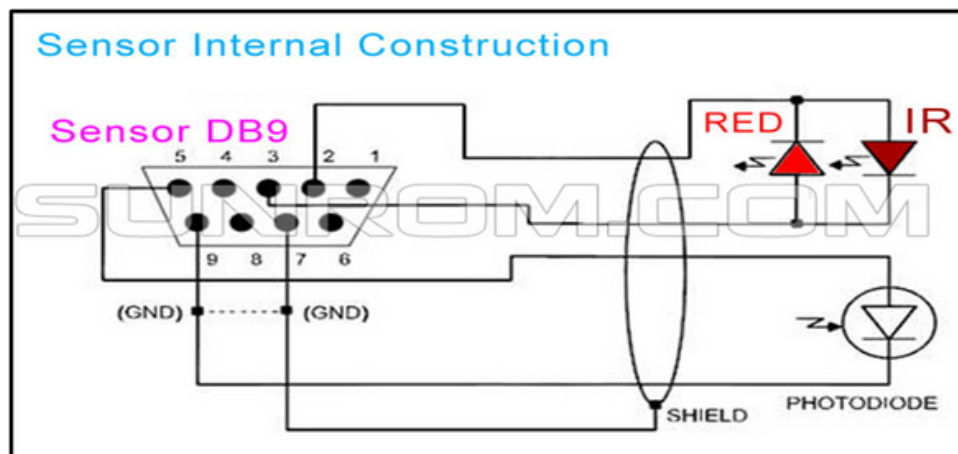
- On the opposite side (or next to the LEDs in reflective mode) is a photodiode that detects the transmitted/reflected light intensity.

Principle of Measurement:

- Oxygenated hemoglobin (HbO₂) and deoxygenated hemoglobin (Hb) absorb red and infrared light differently:

- HbO_2 absorbs more infrared light and less red light.
- Hb absorbs more red light and less infrared light.
- By comparing how much of each wavelength is absorbed during the pulse, the sensor calculates oxygen saturation.

Electronics behind SpO2 Sensor



SpO2 Sensor for pulse oximetry

Connector (Sensor DB9)

- This is a 9-pin connector (DB9 type) used to interface the sensor with external electronics.
- It provides connections for power, ground, LED drive signals, and photodiode output.

Light Sources (LEDs)

- RED LED (~660 nm) → Emits red light.
- IR LED (~940 nm) → Emits infrared light.
- These LEDs shine light through the skin (usually the finger, wrist, or in a ring, the base of the finger).

Light Path Through Tissue

- The LED light passes through blood vessels and tissue.
- Blood absorbs red and infrared light differently, depending on the oxygen level in hemoglobin.

Photodiode (Detector)

- Positioned opposite or adjacent to the LEDs.
- Detects how much light comes through or is reflected back.
- Output is an analog current signal proportional to light intensity.

Shielding

- The "SHIELD" shown in the diagram prevents noise, interference, and stray light from affecting the readings.

Proximity Sensor

What is a proximity sensor?

A proximity sensor detects the presence of an object without physical contact.

- If it is a capacitive sensor, it works by detecting changes in capacitance caused by a nearby object (like your finger or skin).
- Basically, your body (skin, finger, hand) acts as a conductor and changes the electric field around the sensor.

How does a Proximity Sensor work?

A capacitor stores electrical energy between two conductive plates, separated by a dielectric (non-conductive material like air).

Capacitance depends on:

- Plate area
- Distance between plates
- Dielectric constant of the medium

$$C = \epsilon \frac{A}{d}$$

In a Capacitive Sensor

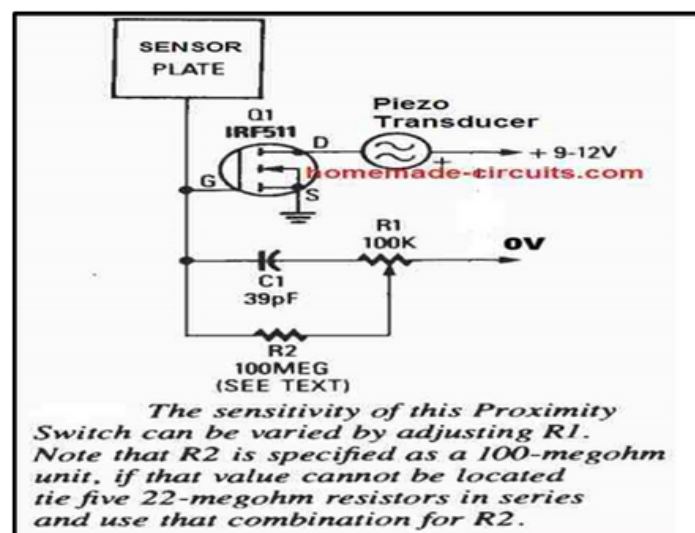
- One "plate" is the sensor electrode inside the ring.
- The other "plate" is your finger/skin or the surrounding environment.

- When your finger comes near, the capacitance changes (because the dielectric medium changes from air → skin/tissue, which has higher permittivity).

Electronics Processing

- The sensor circuit measures these small capacitance changes.
- A microcontroller in the ring interprets this as touch, proximity, or presence detection.

Electronics behind a Proximity Sensor



Proximity Sensor

Sensor Plate

- This is just a piece of conductive material (metal plate or foil).
- It acts as one plate of a capacitor, while the human body (or nearby object) becomes the other plate.
- When a hand comes close, the capacitance changes.

MOSFET (Q1 - IRF511)

- Used as the main amplifier/switch.
- Its Gate (G) is very sensitive to voltage changes caused by capacitance variations at the sensor plate.

- Small variations in capacitance → tiny voltage changes at the gate → large signal amplification at the drain.

R1 (100kΩ)

- This sets the biasing of the MOSFET and helps control sensitivity.
- By adjusting R1, the sensitivity of detection changes.

C1 (39pF)

- Small capacitor that stabilizes the input and sets the working frequency of the oscillation.
- Works with R2 to define response.

R2 (100MΩ)

- Extremely high resistance, ensuring the MOSFET gate has very little leakage current.
- Keeps the circuit sensitive to even tiny capacitance changes.
- If unavailable, multiple 22MΩ resistors in series can be used to reach ~100MΩ.

Microcontroller

- Instead of driving a piezo buzzer, the signal is processed by a microcontroller → used for gesture recognition, touch input, or wake-up function.

POWER COMPONENTS

Rechargeable Battery

What is a Rechargeable Battery?

A rechargeable battery is an electrochemical device that stores energy in chemical form and releases it as electricity when needed. Unlike disposable (primary) batteries, it can be recharged multiple times using an external power source.

In smart rings, the most common type is a Lithium-ion (Li-ion) or Lithium-polymer (Li-Po) battery, because:

- They are small and lightweight.
- They provide high energy density (a lot of energy in a small space).
- They support hundreds of charge cycles.

How does a Rechargeable Battery work?

Energy Storage (Charging)

- When you place the smart ring on a charging dock (usually wireless inductive charging), current flows into the battery.
- Lithium ions move from the cathode (positive electrode) to the anode (negative electrode), where they are stored in a carbon/graphite layer.

Energy Release (Discharging during use)

- When the ring is in use, the ions move back from the anode → cathode, releasing electrons through the external circuit.
- This powers the sensors (PPG, SpO₂, accelerometer, gyroscope, proximity sensor), microcontroller, and Bluetooth module.

Battery Management

A Battery Management System (BMS) inside the ring ensures safe operation. It handles:

- Overcharge protection
- Over-discharge protection
- Temperature monitoring
- Charging cut-off when full



LiPo Battery

In the Context of a Smart Ring

- The rechargeable battery is tiny (10–40 mAh, depending on design).
- It provides 1–3 days of use depending on features enabled (heart-rate tracking, SpO₂, motion sensors, Bluetooth usage).
- The ring uses low-power components so the battery doesn't drain quickly.
- Wireless charging (Qi standard or custom coil) is used to avoid charging ports (since the ring is waterproof/sweatproof).

Wireless Charging Coil

What is a Wireless charging coil?

A wireless charging coil in a smart ring is a tiny inductive coil (a copper wire wound into a loop) that enables contactless power transfer.

- It's part of an inductive charging system (like in phones, smartwatches, and electric toothbrushes).
- The ring has a receiver coil, and the charging dock has a transmitter coil.
- Together, they use electromagnetic induction to transfer energy without physical connectors (keeps the ring waterproof and compact).

How does a Wireless charging coil work?

The process relies on Faraday's law of electromagnetic induction:

"A changing magnetic field in one coil induces a voltage in another nearby coil."

Power Transmission (Dock side)

- The charging dock (base station) has a transmitter coil powered by AC current.
- This creates an oscillating magnetic field around the coil.

Power Reception (Ring side)

- The smart ring has a tiny receiver coil inside.
- When placed near the dock, the magnetic field induces an AC voltage in the ring's coil.

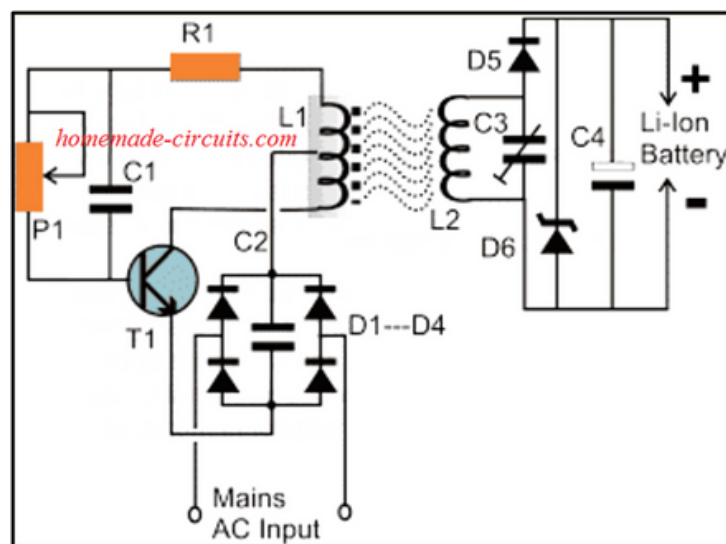
Conversion to Usable Power

- The induced AC voltage is converted into DC voltage using a rectifier circuit.
- A charging IC regulates this DC power and charges the Li-ion/Li-Po battery inside the ring.

Battery Management

- The Battery Management System (BMS) ensures safe charging by monitoring voltage, current, and temperature.

Electronics behind Wireless Charging Coil



Wireless Li-Ion Battery Charger Circuit

Power Input Stage (AC to DC conversion)

- Mains AC Input → D1-D4 (Bridge Rectifier) → C2
- The AC mains (220V/110V) is first converted to DC by the bridge rectifier (D1-D4).
- C2 smooths the rectified DC to provide a stable input.

Oscillator and Primary Coil (Transmitter)

- T1 (Transistor), R1, P1, C1, and L1 form a high-frequency oscillator circuit.
- The oscillator drives L1 (Primary Coil), which generates an alternating magnetic field.

- P1 (Potentiometer) adjusts the frequency and strength of oscillation.
- This magnetic field is the key to wireless energy transfer.

Coupling via Induction

- The magnetic field from L1 induces an AC voltage in L2 (Secondary Coil) through electromagnetic induction.
- This is the wireless power transfer step – no physical wires between primary (charger) and secondary (device).

Rectification and Filtering (Receiver Side)

- The induced AC in L2 is rectified by D5 and D6 (Diodes).
- C3 and C4 act as filter capacitors, smoothing the rectified signal into DC.

Battery Charging

- The output DC is fed to the Li-Ion battery for charging.
- The circuit ensures the voltage is within the safe charging range (~4.2V for a single Li-Ion cell).

In the context of a Smart Ring

- The charging pad (base station) contains the transmitter coil (L1) and oscillator.
- The ring has a small receiver coil (L2) connected to the rectifier and battery.
- When the ring is placed on the charging dock, wireless energy transfer occurs, charging the ring's battery.
- In short: AC mains → Rectifier & Oscillator → Primary coil (L1) → Induction → Secondary coil (L2) → Rectifier & Filter → Li-Ion Battery Charging.

Power Management IC

What is a Power Management IC?

- A Power Management IC is a specialized chip that manages how electrical power is distributed, converted, and stored inside a device.
- In a smart ring, the PMIC acts as the brain of the battery and charging system – making sure the rechargeable battery, wireless charging coil, sensors, and microcontroller all get stable, safe power.

How does a Power Management IC work?

Battery Charging Control

When you place the ring on its wireless charging dock, the charging coil passes power to the PMIC.

The PMIC regulates this incoming energy, ensuring the Li-ion/Li-Po battery charges safely:

- Prevents overcharging (stops charging at ~4.2 V).
- Handles constant-current / constant-voltage (CC/CV) charging.
- Provides temperature monitoring for safety.

Voltage Regulation

The battery voltage (3.7–4.2 V) is not always suitable for all electronics.

The PMIC has DC-DC converters or LDOs (Low Dropout Regulators) that convert battery power into different voltages:

- 1.8 V for sensors (like heart-rate or capacitive sensors).
- 3.3 V for microcontroller / Bluetooth chip.
- ~2.5–3.7 V for haptic motors or LEDs.

This ensures each component gets the exact voltage it needs without wasting power.

Power Switching & Distribution

The PMIC decides which parts of the ring are powered at which time.

Example:

- If the ring is idle, it turns off sensors and keeps only the microcontroller in low-power mode.
- When you move or tap the ring, it wakes up sensors + Bluetooth.
- This is called dynamic power management.

Battery Protection

The PMIC protects the ring from:

- Over-discharge (battery voltage falling too low, which can damage it).
- Over-current (too much power drawn at once).
- Short-circuit (safety shutdown).

Monitoring & Communication

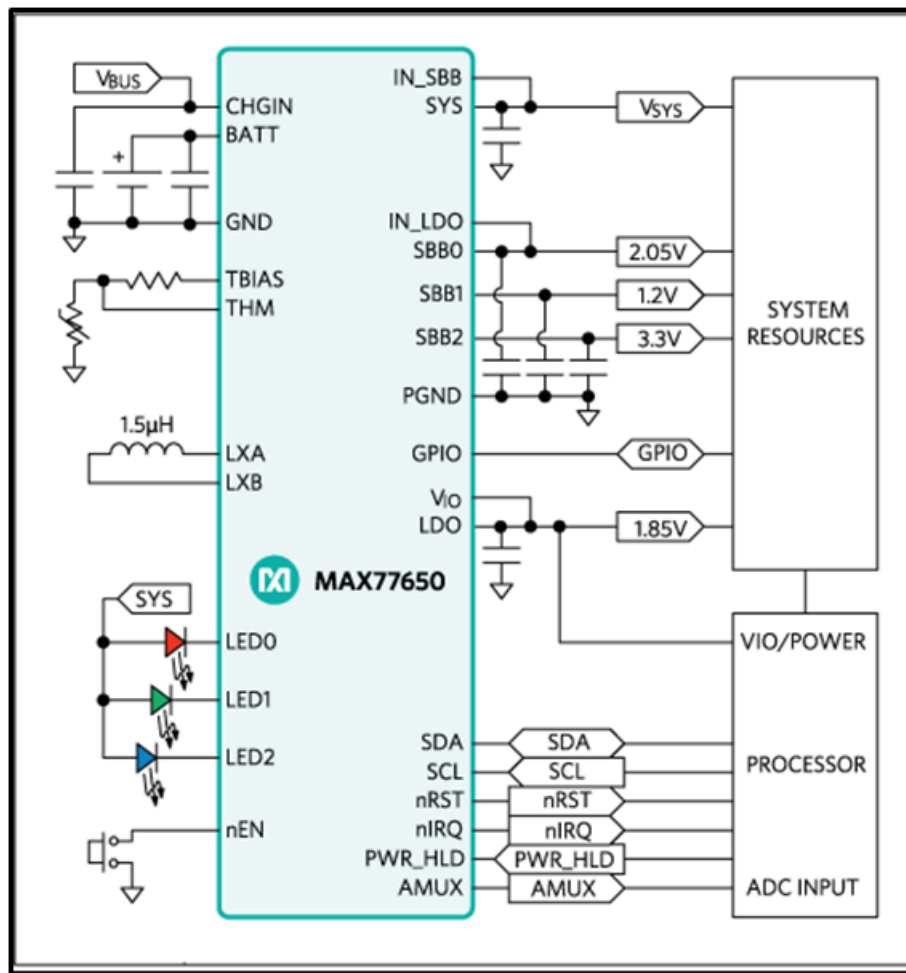
Many PMICs include fuel gauge circuits, which measure:

- Battery level (%).
- Charging/discharging current.
- Health and cycle count.

The PMIC can send this info to the microcontroller, which then shows the battery percentage in the mobile app.

Electronics behind Power Management IC

The MAX77650 PMIC is designed to manage power for small battery-operated devices.



Power Management IC MAX77650

It takes power from a USB charger (Vbus) or a Li-ion rechargeable battery and distributes stable voltages (like 3.3V, 1.8V, 1.2V, etc.) to different subsystems (processor, sensors, LEDs, etc.).

Power Input Section (Top Left)

Vbus / CHGIN / BATT:

- Accepts input from USB charging (Vbus) or battery.
- A battery charger circuit ensures safe charging of the Li-ion battery.
- THM/TBIAS: Monitors battery temperature using an NTC thermistor (safety check during charging).

Switching Regulators (Middle Section)

LXA, LXB, inductor (1.5µH) → Buck Converters (SBB0, SBB1, SBB2):

- Convert battery voltage (~3.7V) down to lower voltages.
- Example outputs: 1.2V, 1.85V, 3.3V for processor, sensors, and memory.

LDOs (IN_LDO, Vio_LDO):

- Provide very stable, noise-free power for sensitive analog circuits (like sensors, ADCs).

System Resources (Top Right)

·Distributes power to different system parts:

- Processor (MCU / SoC) – needs 1.2V/1.8V.
- Sensors – need clean supply (2.05V or 3.3V).
- GPIO outputs for controlling small circuits.

LED Driver (Bottom Left – LED0, LED1, LED2)

- Drives RGB LEDs for notifications, indicators, or sensor illumination (like PPG heart-rate sensors in smart rings).

I²C Interface (Bottom Right – SDA, SCL)

- The processor communicates with the PMIC via I²C
- The processor can turn on/off rails, monitor battery level, or control LEDs.

Other Control Signals

- nEN: Enable pin for the chip.
- PWR_HLD: Power-hold signal to keep the system alive.
- nIRQ: Interrupt pin (alerts the processor about events like low battery).
- nRST: Reset pin to restart system power sequencing.

In the Context of a Smart Ring

Charging

- When you place the smart ring on a wireless charging pad, the charging coil + rectifier provides DC to CHGIN, charging the Li-ion battery safely.

Power Regulation

The PMIC converts the battery voltage (3.0–4.2V) into different regulated voltages:

- 3.3V → for Bluetooth chip & sensors.
- 1.8V / 1.2V → for microcontroller or memory.
- Stable LDO output → for analog sensors.

Monitoring

- The PMIC tracks battery voltage, current, and temperature.
- Sends alerts (low battery, charging complete) to the microcontroller via I²C & IRQ pin.

LED Control

- The PMIC directly drives indicator LEDs (like charging status or heart rate sensor LED).

COMMUNICATION MODULES

Bluetooth Low Energy

What is Bluetooth Low Energy?

- Bluetooth Low Energy (BLE) is a wireless communication protocol designed for short-range data exchange with minimal power consumption.
- Unlike "classic" Bluetooth (used for headphones, speakers, etc.), BLE is optimized for devices that need to send small amounts of data intermittently while running on tiny batteries (like in a smart ring).
- BLE typically operates in the 2.4 GHz ISM band and supports ranges from a few meters up to 100+ meters (depending on environment and antenna design).

How does Bluetooth Low Energy work?

Advertising and Scanning

- The smart ring (BLE peripheral) sends out advertising packets (short bursts of data) at intervals.
- A smartphone (BLE central device) scans for these packets.
- This uses very little power because the ring is not transmitting continuously.

Connection Establishment

When your phone detects the smart ring, it can establish a BLE connection.

This connection allows secure two-way communication:

- Smart ring → Phone: Sends sensor data (heart rate, movement, temperature, etc.).
- Phone → Smart ring: Sends commands (e.g., firmware updates, settings changes, notification alerts).

GATT Protocol (Generic Attribute Profile)

BLE uses the GATT profile to organize data into services and characteristics.

Example for a smart ring:

- Heart Rate Service → Sends beats per minute (BPM).
- Battery Service → Sends battery percentage.
- Motion Service → Sends activity/gesture data.

Each service is like a "folder," and each characteristic is like a "file" (data value).

Low Power Operation

BLE devices (like smart rings) sleep most of the time and wake up only when:

- They need to send a data packet.
- The smartphone requests an update.

This way, the ring can run for days or even weeks on a tiny Li-ion battery (20–50 mAh).

Example in a Smart Ring

Imagine a fitness-focused smart ring:

Sensors: Heart rate monitor, accelerometer, temperature sensor.

BLE role:

- Periodically sends heart rate and activity data to your phone.
- Notifies you of phone calls/messages via vibration or LED.

- Receives configuration updates (e.g., enable/disable certain sensors).
- Power savings: The ring sends data only every few seconds/minutes instead of continuously, saving battery.

In the Context of a Smart Ring

Power Input

- The rechargeable Li-ion battery in the smart ring powers the nRF52840 through its regulated supply pins.
- When charging (via wireless charging coil), power also flows to the chip.

Sensor Data Acquisition

- Sensors like PPG/SpO₂, EDA, accelerometer, and gyroscope are connected via I²C or SPI pins.
- The chip reads raw signals (light absorption, motion, skin conductance, etc.).

Data Processing

The ARM Cortex-M4F CPU inside the nRF52840 processes the raw sensor data into meaningful metrics like:

- Heart rate
- Blood oxygen (SpO₂)
- HRV (Heart Rate Variability)
- Sleep stages
- Steps, activity tracking

Bluetooth Communication

- Processed data is sent wirelessly via the BLE 5.0 radio to a smartphone app.
- Example: Your ring streams live HRV data to the Ultrahuman/Oura app.

Low Power Management

- The 32.768 kHz crystal enables deep sleep while still keeping track of time.
- The chip only wakes periodically (or when motion/EDA triggers) → extending battery life.

User Interaction

- GPIOs can drive tiny LEDs for notifications or be used in the PPG sensor module (red/IR/green LEDs).
- Touch/proximity sensors also feed signals back to the chip for user input.

NFC Chip

What is an NFC Chip?

NFC chips in smart rings are the tiny integrated circuits that let the ring communicate wirelessly over very short distances (usually < 4 cm).

They enable features like:

- Payments (like Apple Pay, Google Pay, or bank-issued NFC cards).
- Access Control (unlocking smart locks, office doors, hotel rooms).
- Transit Passes (metro/train cards).
- Identity / authentication (logging into devices, 2FA).

How does an NFC chip work?

Idle State

- The NFC chip in the smart ring doesn't need battery power.
- It stays passive, waiting for an external NFC reader (payment terminal, door lock, transport gate).

Activation

- When you bring the ring close to the NFC reader (within ~4 cm), the reader sends out a 13.56 MHz electromagnetic field.
- The antenna coil in the ring captures this energy and powers the NFC chip.

Handshake & Communication

- The NFC chip establishes a secure communication link with the reader.
- The chip verifies the request (e.g., “payment” or “door unlock”).

Secure Data Exchange

- If it's a payment → The chip doesn't send your actual card number.
- Instead, it sends a one-time cryptographic token stored in the secure element.
- The reader forwards this to the bank's system for validation.

Completion

- The transaction or unlock happens in less than a second.
- The NFC chip then goes back to idle mode, ready for the next tap.

Electronics behind NFC chip

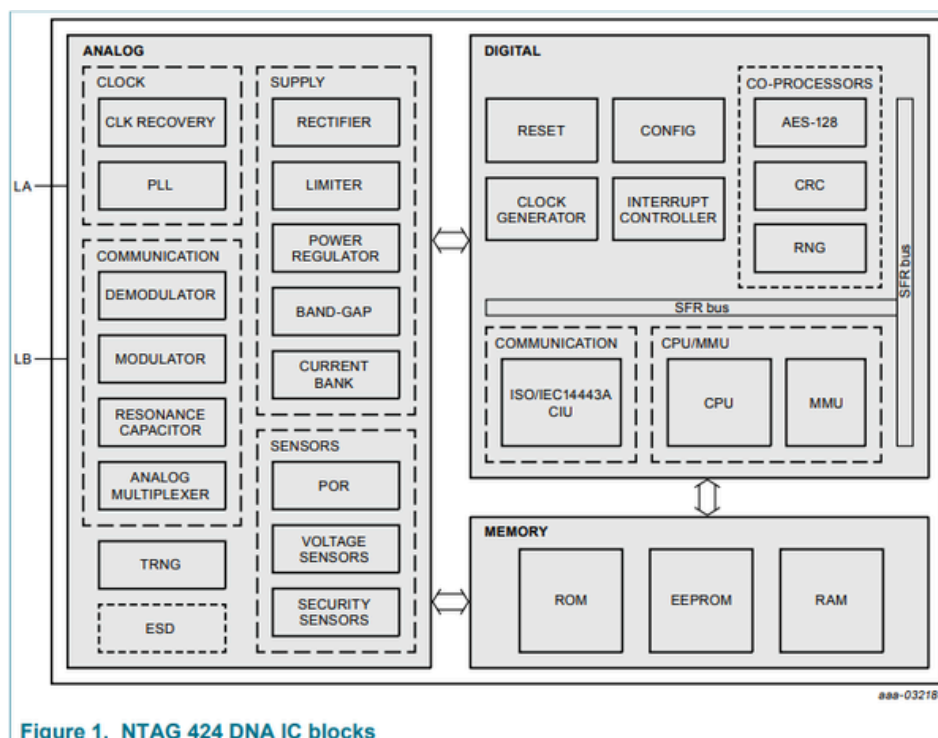


Figure 1. NTAG 424 DNA IC blocks

NFC Chip block diagram

Power & Communication

- NFC is passive in most smart rings.
- The ring has no big battery – instead, it draws power from the NFC reader's electromagnetic field (like a payment terminal or phone).
- The analog block in the diagram:
 - Resonance Capacitor + Rectifier + Power Regulator → capture RF energy from the reader's field, convert it into DC power.
 - Demodulator/Modulator → handle data exchange with the reader by modulating the RF field.

Data Processing & Security

- The digital block (CPU, MMU, interrupt controller) processes commands from the NFC reader.
- Security features:
 - AES-128 Crypto Engine → ensures secure transactions (e.g., payment, access control).
 - CRC & RNG → error detection and generation of secure random numbers for authentication.
 - Security Sensors → protect against tampering (voltage glitches, temperature attacks).

Memory Usage

- EEPROM: Stores keys, user credentials (e.g., payment tokens, ID).
- ROM: Holds the fixed firmware.
- RAM: Temporary workspace during communication.

Transaction Flow in a Smart Ring

- You tap your smart ring to an NFC reader (POS terminal, door lock, smartphone).
- The ring's antenna picks up the RF field → powers the NFC chip.
- The chip authenticates itself using AES-based challenge-response protocols.

- If verified, it shares the necessary credential (like a tokenized card number).
- The reader processes the transaction → payment or access granted.

In short:

The NFC chip in a smart ring (like NTAG 424 DNA) powers itself wirelessly, securely authenticates with the reader, and shares stored credentials from EEPROM, enabling contactless payments, access, or ID functions.

MICROCONTROLLERS

What is a Microcontroller?

A microcontroller is a small, compact integrated circuit that functions as a self-contained computer designed for specific tasks within embedded systems. It typically includes a processor core (CPU), memory (both for storing programs and data), and input/output peripherals all on a single chip. The microcontroller interprets and processes data from inputs such as sensors, then controls outputs to devices like motors or displays based on programmed instructions.

How does a Microcontroller work?

Initialization:

When the smart ring is powered on, the microcontroller initializes its internal components and peripherals such as sensors (heart rate, accelerometer, gyroscope), memory, and Bluetooth module.

Data Acquisition:

The microcontroller continuously collects raw data from integrated sensors. For example, it reads signals from the optical heart rate sensor, motion sensors, and temperature sensors through interfaces like analog-to-digital converters or digital buses.

Data Processing:

It processes the raw sensor data locally using embedded algorithms to filter noise and extract meaningful metrics like heart rate, step count, or sleep quality without sending all raw data to external devices.

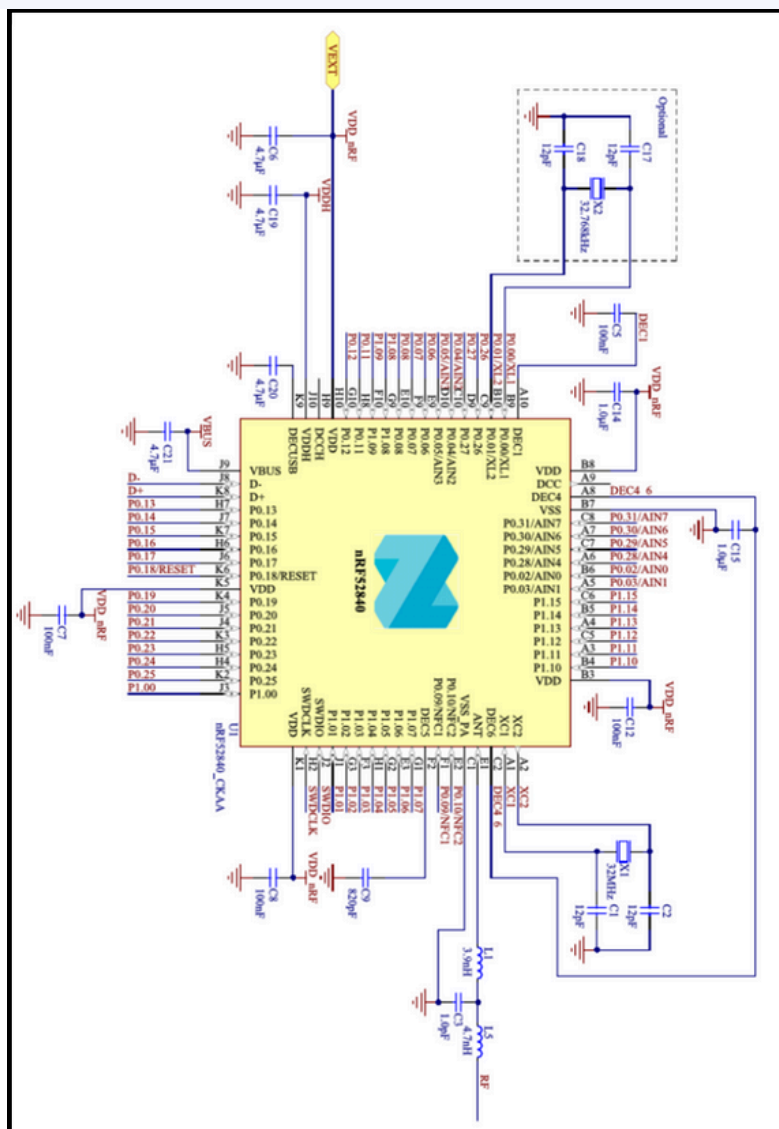
Power Management:

To conserve battery life in the limited space of a ring, the microcontroller controls power states of the sensors and wireless modules. It uses techniques like sleep modes, selective power gating, and dynamic frequency scaling to maximize operational time.

User Interaction:

It controls any input/output features in the ring, such as haptic feedback for notifications, LED indicators, or capacitive touch sensing, providing a user interface.

Electronics behind a Microcontroller



Core Processor:

64 MHz 32-bit Arm Cortex-M4 CPU with Floating Point Unit (FPU) for high-performance and efficient computation, including DSP and floating-point math capabilities.

Memory:

1 MB Flash memory for program storage and 256 KB RAM for data and stack usage, supporting complex applications and multitasking.

Wireless Connectivity:

Multiprotocol 2.4 GHz radio supporting:

- Bluetooth 5.3 Low Energy (LE) with features such as 2 Mbps data rate, long range, advertising extensions, and channel selection algorithm #2.
- IEEE 802.15.4 standard for Thread and Zigbee protocols.
- ANT protocol and proprietary 2.4 GHz RF.

Radio Performance:

- Transmit power programmable from +8 dBm to -20 dBm.
- Sensitivity down to approximately -103 dBm depending on data rate.

Security:

Integrated Arm CryptoCell CC310 cryptographic accelerator supporting AES, ECC, and SHA cryptographic operations for advanced, hardware-based security.

Interfaces:

- USB 2.0 full speed (12 Mbps) interface.
- High-speed SPI (32 MHz) and Quad SPI (QSPI) for connecting external memory/display.
- TWI (I2C), I²S, PDM, PWM, and more for wide peripheral connectivity.
- 12-bit ADC for analog sensor data.
- NFC-A tag support for simplified pairing and data transfer.

Additional Features:

EasyDMA for offloading peripherals and improving performance, QSPI for fast external flash interfacing, and multiple security and communication protocols integration.

In the Context of a Smart Ring

The Nordic Semiconductor nRF52840 microcontroller plays a crucial role by providing the necessary processing power, wireless connectivity, and security features tailored for compact wearable devices.

Size and Integration:

The nRF52840 comes in a wafer-level chip-scale package (WLCSP), making it small enough to fit comfortably inside the tiny form factor of a smart ring, typically around 8 mm wide and 2.9 mm thick.

Sensor Supervision and Data Processing:

- The 64 MHz ARM Cortex-M4 processor with a floating point unit efficiently supervises various health sensors embedded in the ring, such as heart rate monitors, blood oxygen sensors (SpO2), temperature sensors, and motion detectors.
- It processes this raw sensor data locally using sophisticated algorithms to deliver real-time insights like heart rate variability, sleep quality, and activity levels.

Wireless Connectivity:

The integrated Bluetooth Low Energy (BLE) radio allows the smart ring to maintain a stable and energy-efficient wireless connection to smartphones or other devices. This connectivity enables continuous syncing of health data to apps for visualization, personalized advice, and goal-setting.

Power Efficiency:

Battery life is critical in a small device, and the nRF52840's ultra-low power design, including automated power management and low peak current consumption (around 5.5 mA for Bluetooth operations), enables the ring to last up to 10 days on a single charge or up to 30 days in standby mode.

Security:

The built-in Arm CryptoCell-310 hardware accelerator ensures that sensitive personal health data collected by the smart ring is encrypted and securely transmitted, protecting user privacy.